

LEVERAGED BASIS STRATEGY

Backtest Report · Jan 2022 – Mar 2026

+488% Total Return	+52.1% CAGR	4.62 Sharpe Ratio	-23.0% Max Drawdown
\$58,794 \$10k → Final	+42.3% Net APR (avg)	8.03 Sortino Ratio	2.27 Calmar Ratio

Strategy: 5× ETH Loop · 80% LTV/loop · ×3.36 leverage · Delta-Neutral via ETH-USDT Perp Short
· stETH staking 2.7%

Prepared: 26 March 2026 · Confidential

Strategy Overview

What is it?

The Leveraged Basis Strategy is a **delta-neutral carry trade** on Ethereum. It earns yield from two independent sources — perpetual funding rates and stETH staking rewards — while remaining fully hedged against ETH price movements.

How it works

- 1. Collateral loop (5×):** Starting with 1 stETH, the strategy repeatedly deposits stETH, borrows USDC at 80% LTV, buys more stETH, and re-deposits. After 5 loops this creates a ×3.36 leveraged stETH position on-chain.
 - 2. Delta hedge:** An equal-sized short position on the ETH-USDT perpetual futures (Binance) cancels all directional ETH exposure. The portfolio is **price-neutral** — it neither gains nor loses from ETH price moves.
 - 3. Revenue streams:**
 - **Funding income:** Perpetual longs pay shorts when the market is bullish (avg raw rate 15.4% APR). At ×3.36 leverage this amplifies to ~46% APR on equity.
 - **Staking income:** All looped stETH earns ~2.7% APR, amplified to ~9.1% APR on equity.
 - 4. Cost:** USDC borrow interest on the leveraged position (avg 5.6% APR including 0.5% protocol spread). This nets to ~13% APR drag on equity.
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Quarterly rebalancing

At the end of each quarter the position is rebalanced: collateral, short notional, and borrow size are all rescaled to the current portfolio value. This ensures the leverage ratio stays constant and compounding gains are fully deployed.

Key risks

- **Negative funding:** When the market turns bearish, longs pay nothing and shorts may pay longs (2022Q3: -53.6% annualised; 2025Q1: -35.5%).
 - **Borrow rate spikes:** DeFi utilisation surges can push USDC rates above funding income (notable in late 2023 when borrow cost hit ~20% annualised).
 - **Liquidation risk:** 80% LTV per loop leaves a buffer, but extreme price volatility or oracle manipulation could threaten collateral ratios.
 - **Smart-contract risk:** Aave/Compound protocol failures or exploits.
 - **Execution slippage:** Modelled at 10% funding take-rate.
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Backtest assumptions

Period: Jan 2022 – Mar 2026 (4.2 years) · Loops: 5 · LTV: 80%/loop · Leverage: ×3.36 · Funding take-rate: 10% · Borrow spread: +0.5% · Staking APR: 2.7% · Data: Binance ETH-USDT 8h funding; Aave/Compound USDC borrow; CoinGecko ETH price.

Risk Statistics

+487.9% Total Return	+52.1% CAGR	9.2% Ann. Volatility	4.62 Sharpe Ratio
8.03 Sortino Ratio	2.27 Calmar Ratio	-23.0% Max Drawdown	252 days Max DD Duration
-0.620% VaR 95% (daily)	-1.016% VaR 99% (daily)	-0.881% CVaR 95%	-1.303% CVaR 99%
57.8% Win Rate	1.93 Profit Factor	+0.28 Skewness	+0.90 Excess Kurtosis
+1.51% Best Day	-1.52% Worst Day	+29.7% Best Month	-12.4% Worst Month

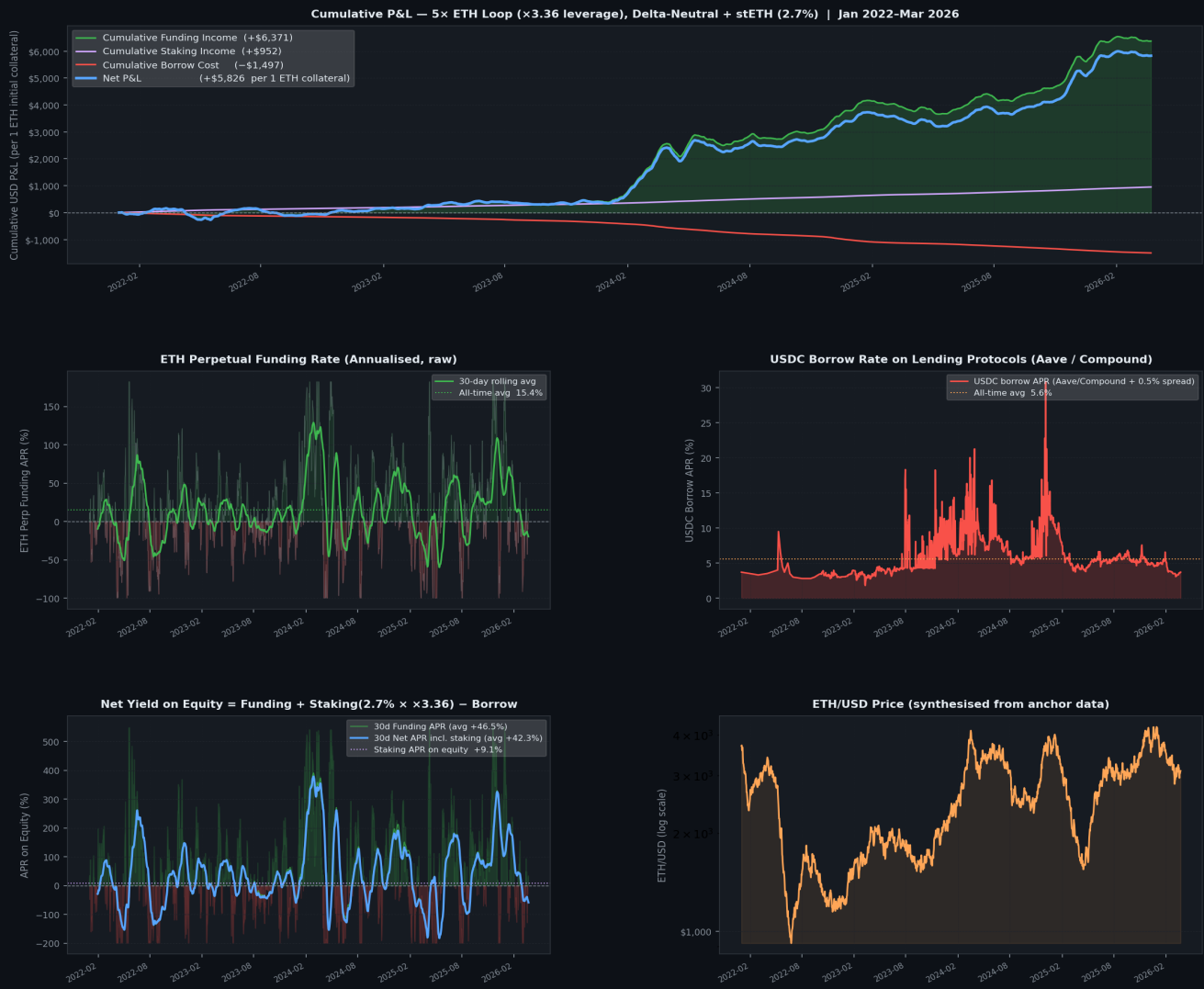
P&L; Attribution (per 1 stETH collateral)

+\$6,371 Total Funding Income	+\$952 Total Staking Income	-\$1,497 Total Borrow Cost	+\$5,826 Total Net P&L;
+46.5% Avg Funding APR	+9.1% Avg Staking APR	-13.2% Avg Borrow APR	+42.3% Avg Net APR

Cumulative P&L; Breakdown

Funding income (green), staking income (purple), borrow cost (red) and net P&L; (blue) — all cumulative USD per 1 stETH initial collateral.

Leveraged ETH Basis Backtest | Jan 2022 - Mar 2026



Summary Statistics

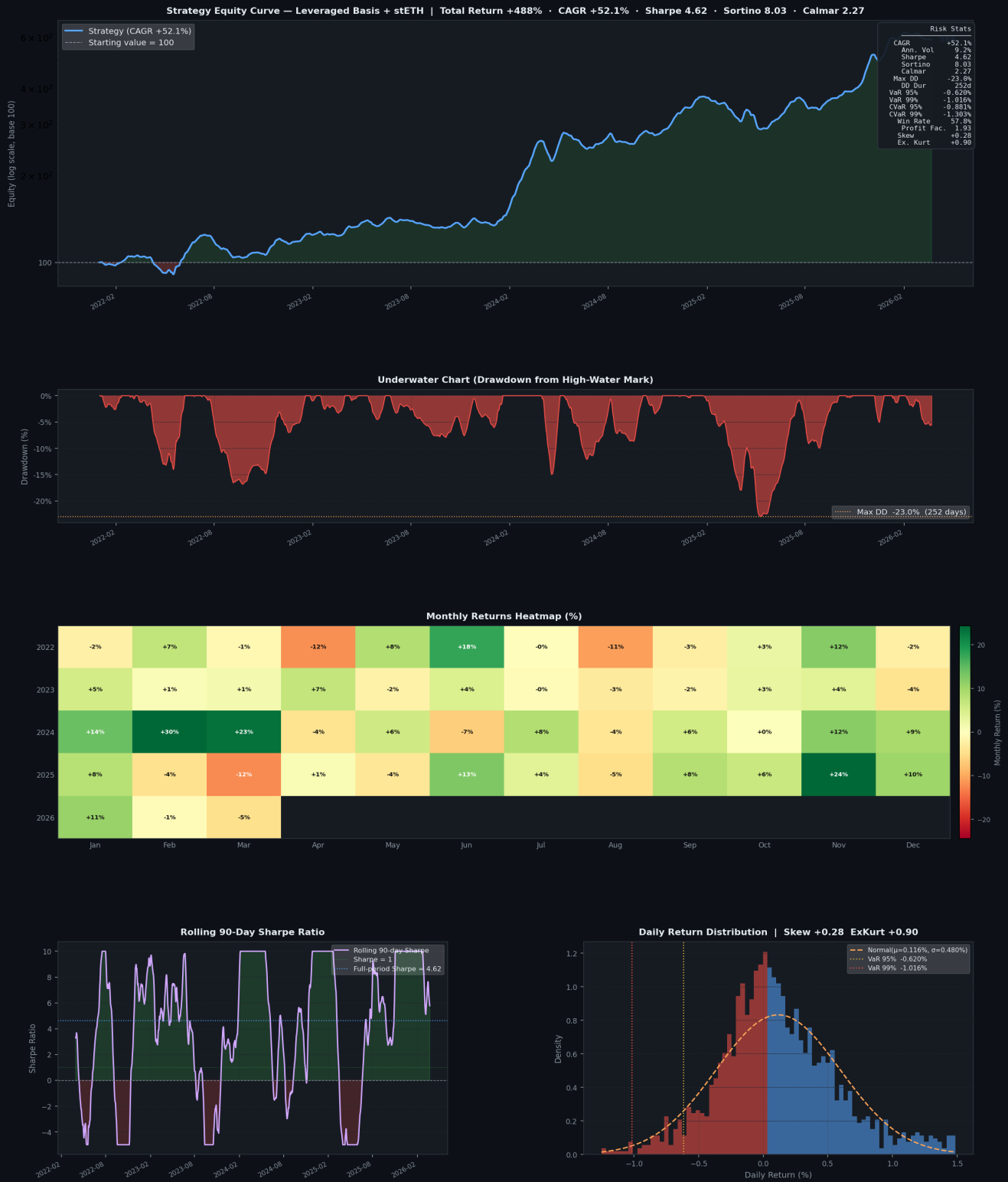
Strategy: 5x ETH Loop · 80% LTV/loop · x3.36 leverage · Delta-Neutral via ETH-USDT Perp Short

Period:	Jan 2022 - Mar 2026 (4.2 yrs)	Total Net P&L:	+\$5,826 per ETH	Sharpe Ratio:	4.62
Initial collateral:	1 stETH (~\$3,700)	Avg Net Yield (on equity):	+42.3% APR	Max Drawdown (USD):	\$-532
Total Funding Income:	+\$6,371 per ETH	Avg Funding APR on equity:	+46.5% APR	% Days Profitable:	57.8%
Total Staking Income (2.7%):	+\$952 per ETH	Avg Staking APR on equity:	+9.1% APR		
Total Borrow Cost:	-\$1,497 per ETH	Avg Borrow Cost on equity:	-13.2% APR (cost)		

Strategy Performance & Risk Analysis

Equity curve (log scale), underwater drawdown chart, monthly return heatmap, rolling 90-day Sharpe ratio, and daily return distribution.

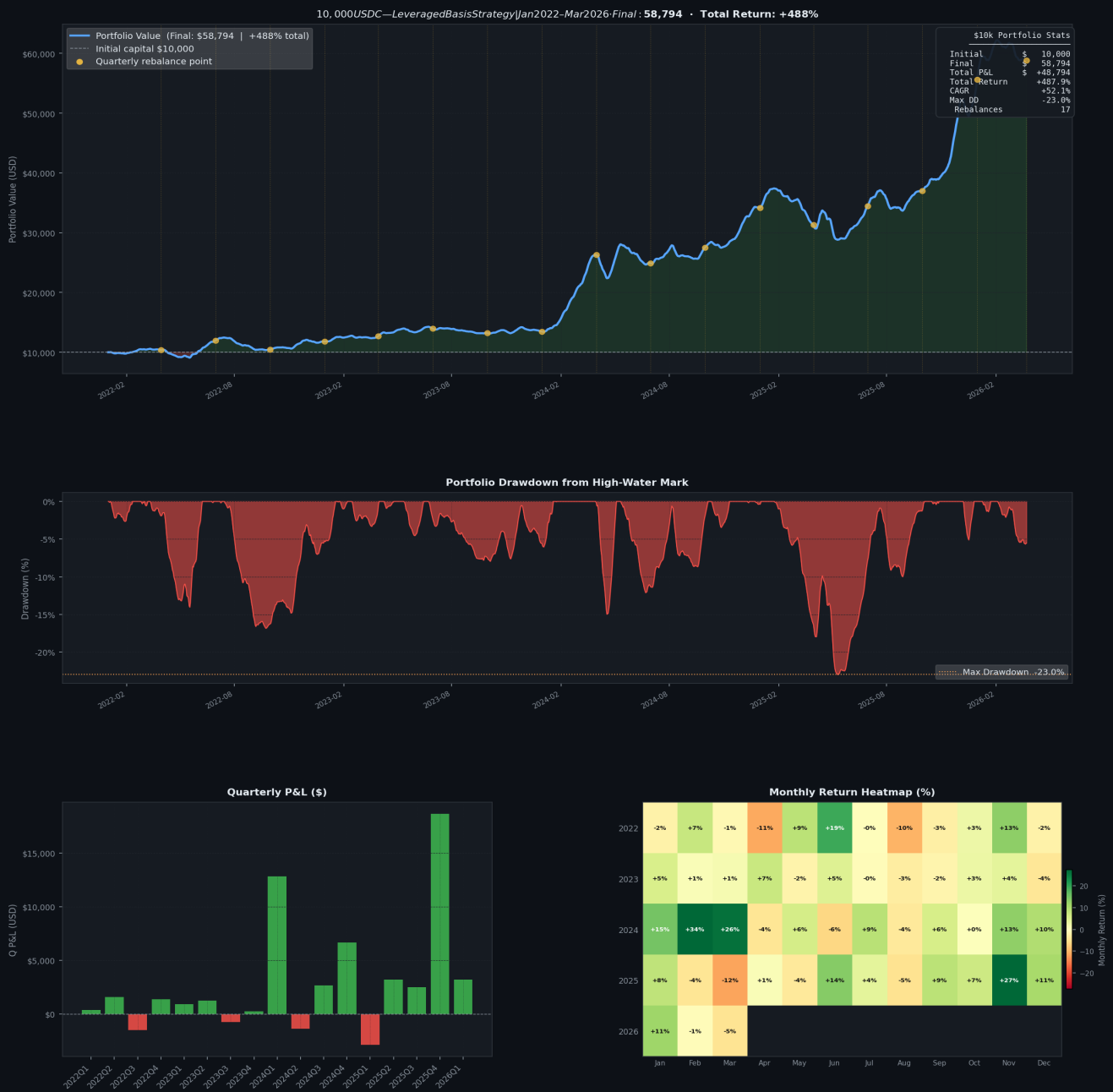
Leveraged Basis Strategy — Performance & Risk Analysis | Jan 2022 - Mar 2026



\$10,000 USDC Portfolio Simulation

Starting with \$10,000 USDC, daily compounding of net yield, position rebalanced at each quarter-end. Final value: **\$58,794** · Total return: **+488%** · CAGR: **+52.1%**

\$10,000 USDC — Leveraged Basis Strategy Portfolio | Jan 2022 - Mar 2026



Quarterly Breakdown

Annualised % return on equity per quarter (per 1 stETH collateral). Borrow cost shown as negative.

Quarter	Funding APR	Staking APR	Borrow Cost	Net APR	Gross Income	Q Return (\$10k)	Days
2022Q1	+14.2%	+9.1%	-8.2%	+15.1%	+23.3%	+3.8%	90
2022Q2	+59.0%	+9.1%	-10.4%	+57.7%	+68.1%	+15.3%	91
2022Q3	-55.8%	+9.1%	-6.9%	-53.6%	-46.8%	-12.7%	92
2022Q4	+47.6%	+9.1%	-7.7%	+49.0%	+56.7%	+13.1%	92
2023Q1	+28.8%	+9.1%	-7.8%	+30.2%	+37.9%	+7.7%	90
2023Q2	+37.4%	+9.1%	-9.1%	+37.5%	+46.5%	+9.7%	91
2023Q3	-17.6%	+9.1%	-13.2%	-21.7%	-8.5%	-5.3%	92
2023Q4	+16.1%	+9.1%	-17.3%	+7.9%	+25.1%	+2.0%	92
2024Q1	+283.8%	+9.1%	-23.4%	+269.5%	+292.9%	+95.1%	91
2024Q2	-8.7%	+9.1%	-21.0%	-20.6%	+0.4%	-5.2%	91
2024Q3	+45.7%	+9.1%	-14.4%	+40.4%	+54.8%	+10.6%	92
2024Q4	+99.1%	+9.1%	-22.2%	+86.1%	+108.2%	+24.1%	92
2025Q1	-28.8%	+9.1%	-15.7%	-35.5%	-19.8%	-8.5%	90
2025Q2	+42.3%	+9.1%	-11.6%	+39.8%	+51.4%	+10.2%	91
2025Q3	+32.3%	+9.1%	-13.4%	+27.9%	+41.3%	+7.2%	92
2025Q4	+166.1%	+9.1%	-12.3%	+162.9%	+175.2%	+50.4%	92
2026Q1	+25.7%	+9.1%	-10.0%	+24.7%	+34.7%	+5.7%	83
TOTAL	+46.5%	+9.1%	-13.2%	+42.3%	+55.6%	+487.9%	1544

Monthly Portfolio P&L; — \$10,000 USDC

Daily compounding, position rebalanced at each quarter-end. Negative months highlighted red, positive green.

Month	Start (\$)	End (\$)	P&L (\$)	Return %	Net APR %	Gross APR %	Days
Jan 2022	\$10,000	\$9,774	\$-226	-2.26%	-26.8%	-18.3%	31
Feb 2022	\$9,774	\$10,489	+\$715	+7.32%	+92.2%	+100.3%	28
Mar 2022	\$10,489	\$10,376	\$-114	-1.09%	-12.8%	-4.8%	31
Apr 2022	\$10,376	\$9,209	\$-1,167	-11.25%	-144.7%	-136.0%	30
May 2022	\$9,209	\$10,014	+\$805	+8.75%	+99.6%	+112.7%	31
Jun 2022	\$10,014	\$11,959	+\$1,945	+19.42%	+216.8%	+226.1%	30
Jul 2022	\$11,959	\$11,931	\$-28	-0.24%	-2.6%	+4.3%	31
Aug 2022	\$11,931	\$10,717	\$-1,214	-10.18%	-126.0%	-119.4%	31
Sep 2022	\$10,717	\$10,440	\$-276	-2.58%	-31.7%	-24.5%	30
Oct 2022	\$10,440	\$10,704	+\$264	+2.53%	+29.5%	+37.6%	31
Nov 2022	\$10,704	\$12,080	+\$1,376	+12.86%	+147.7%	+155.4%	30
Dec 2022	\$12,080	\$11,805	\$-275	-2.28%	-27.0%	-19.8%	31
Jan 2023	\$11,805	\$12,451	+\$646	+5.47%	+62.9%	+71.0%	31
Feb 2023	\$12,451	\$12,542	+\$91	+0.73%	+9.6%	+17.7%	28
Mar 2023	\$12,542	\$12,712	+\$171	+1.36%	+16.0%	+23.2%	31
Apr 2023	\$12,712	\$13,623	+\$911	+7.17%	+84.5%	+93.1%	30
May 2023	\$13,623	\$13,340	\$-284	-2.08%	-24.6%	-15.2%	31
Jun 2023	\$13,340	\$13,950	+\$610	+4.57%	+54.6%	+63.8%	30
Jul 2023	\$13,950	\$13,888	\$-62	-0.44%	-5.2%	+7.1%	31
Aug 2023	\$13,888	\$13,460	\$-428	-3.08%	-36.8%	-21.5%	31
Sep 2023	\$13,460	\$13,206	\$-254	-1.89%	-23.1%	-11.2%	30
Oct 2023	\$13,206	\$13,562	+\$356	+2.69%	+31.4%	+44.0%	31
Nov 2023	\$13,562	\$14,065	+\$503	+3.71%	+44.6%	+63.5%	30
Dec 2023	\$14,065	\$13,466	\$-599	-4.26%	-51.1%	-30.8%	31
Jan 2024	\$13,466	\$15,495	+\$2,029	+15.07%	+165.9%	+185.9%	31
Feb 2024	\$15,495	\$20,814	+\$5,319	+34.33%	+373.5%	+395.3%	29
Mar 2024	\$20,814	\$26,273	+\$5,458	+26.22%	+275.6%	+304.2%	31
Apr 2024	\$26,273	\$25,115	\$-1,157	-4.41%	-52.9%	-30.9%	30
May 2024	\$25,115	\$26,581	+\$1,466	+5.84%	+67.7%	+87.3%	31
Jun 2024	\$26,581	\$24,896	\$-1,685	-6.34%	-79.4%	-58.0%	30
Jul 2024	\$24,896	\$27,085	+\$2,188	+8.79%	+99.4%	+116.6%	31
Aug 2024	\$27,085	\$26,041	\$-1,044	-3.85%	-45.8%	-32.6%	31
Sep 2024	\$26,041	\$27,542	+\$1,501	+5.76%	+68.5%	+81.3%	30
Oct 2024	\$27,542	\$27,637	+\$95	+0.35%	+4.2%	+18.9%	31
Nov 2024	\$27,637	\$31,142	+\$3,505	+12.68%	+145.7%	+165.8%	30
Dec 2024	\$31,142	\$34,188	+\$3,046	+9.78%	+110.2%	+141.9%	31
Jan 2025	\$34,188	\$37,037	+\$2,849	+8.33%	+94.5%	+117.0%	31
Feb 2025	\$37,037	\$35,429	\$-1,608	-4.34%	-57.7%	-43.7%	28
Mar 2025	\$35,429	\$31,297	\$-4,131	-11.66%	-145.5%	-135.0%	31

Month	Start (\$)	End (\$)	P&L (\$)	Return %	Net APR %	Gross APR %	Days
Apr 2025	\$31,297	\$31,530	+\$232	+0.74%	+10.1%	+20.0%	30
May 2025	\$31,530	\$30,374	\$-1,155	-3.66%	-43.1%	-31.1%	31
Jun 2025	\$30,374	\$34,492	+\$4,117	+13.56%	+155.1%	+168.1%	30
Jul 2025	\$34,492	\$35,780	+\$1,288	+3.73%	+43.5%	+56.0%	31
Aug 2025	\$35,780	\$34,054	\$-1,726	-4.82%	-57.9%	-44.2%	31
Sep 2025	\$34,054	\$36,977	+\$2,923	+8.58%	+100.4%	+114.5%	30
Oct 2025	\$36,977	\$39,442	+\$2,465	+6.67%	+76.2%	+88.5%	31
Nov 2025	\$39,442	\$50,258	+\$10,817	+27.42%	+296.4%	+309.9%	30
Dec 2025	\$50,258	\$55,615	+\$5,357	+10.66%	+120.4%	+131.7%	31
Jan 2026	\$55,615	\$61,942	+\$6,327	+11.38%	+127.4%	+139.5%	31
Feb 2026	\$61,942	\$61,588	\$-355	-0.57%	-7.4%	+1.9%	28
Mar 2026	\$61,588	\$58,794	\$-2,794	-4.54%	-70.4%	-62.3%	24